Application of NPP-VIIRS Nighttime Light Imagery for GDP Spatialization and Economic Differences in Vietnam

DA 401 Early Results

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Abstract

Gross domestic product (GDP) is a fundamental indicator of economic activity within a country or region. However, GDP statistics are often reported at relatively aggregated administrative levels, making it difficult for researchers and policymakers to examine small-scale patterns of economic development. Nighttime light (NTL) data collected by the Suomi National Polar-orbiting Partnership (Suomi NPP) satellite provide an alternative source of information for estimating economic activity at finer spatial scales. This paper evaluates four predictive paradigms to estimate commune-level GDP per capita in Vietnam using NTL data, while accounting for population and imposing constraints based on known regional GDP totals. The paradigms include Ordinary Least Squares, Ridge Regression, Geographically Weighted Regression, and Neural Network. These methods explain up to 47% of the variation when used at the provincial administrative level, while overestimating known outliers when applied at the commune level. By combining NTL radiance with economic data, this study aims to produce reliable estimations of economic activity across communes in Vietnam, while commune-level self-computation might create a mismatch in fine-level GDP estimates.

1 Introduction

Gross domestic product (GDP) is a fundamental economic indicator within a country or region. More detailed GDP estimates can provide researchers and policymakers with a clearer understanding of regional economic disparities and development patterns. However, conventional socioeconomic statistics used to measure GDP often suffer from inconsistent statistical scales and limited spatial resolution, making it difficult to capture regional economic disparities at a fine spatial scale. Therefore, many studies have proposed various methods for constructing maps and data on regional GDP with respect to real world coordinates. Researchers have shown that *nighttime light* (NTL) data are a useful economic indicator and have adopted and developed methods for spatializing GDP using NTL data, including both statistical and machine learning approaches. However, applications of NTL data for GDP spatialization in Vietnam remain limited, motivating this study to spatialize regional GDP in Vietnam using NTL data.

Remote sensing imagery offers clear advantages over conventional socioeconomic statistics for describing the spatial GDP distribution. Before *Visible Infrared Imaging Radiometer Suite* (VIIRS) data, carried by the *Suomi National Polar-Orbiting Partnership* (NPP), became available in 2013, researchers primarily relied on NTL data from the *Defense Meteorological Satellite Program-Operational Linescan System* (DMSP-OLS). These studies found a strong relationship between

NTL intensity and regional GDP. For example, Elvidge et al. (1997) found a strong correlation between the area lit and GDP across 21 counties in North America using DMSP-OLS NTL data. Similar studies conducted on the data from the United States, Europe, and China have also reported strong relationships between NTL and GDP (Doll et al., 2006; Forbes, 2013; N. Zhao et al., 2011).

Following the introduction of NPP-VIIRS data in 2013, research increasingly shifted toward using this newer dataset (Li et al., 2013; M. Zhao et al., 2017). NPP-VIIRS provides higher spatial resolution (approximately 500 m) than DMSP-OLS data (1 km), while its wider radiometric detection range reduces saturation and its onboard radiometric calibration improves data quality (Elvidge et al., 2013). However, NPP-VIIRS imagery is a preliminary product affected by various sources of light, including cities, transportation corridors, gas flares, biomass burning, background noise, and reflections from bright surfaces such as snow covered mountains and playa lake beds (Elvidge et al., 2013). To address these issues, M. Zhao et al. (2017) proposed a method for correcting VIIRS nighttime light data. Building upon the procedure developed by M. Zhao et al. (2017), Xu et al. (2024) estimated regional GDP for all provinces and prefectures in China while incorporating land cover data. Their results also highlighted the effectiveness of *Geographically Weighted Regression* (GWR) in capturing economic development using NTL data. The GWR model explained up to 93% of the variation in economic development, compared with 82% explained by the *Ordinary Least Squares* (OLS) model (Xu et al., 2024). Liu et al. (2025) used NPP-VIIRS nighttime light and land use data to develop a 1 kmgridded GDP dataset for China using machine learning methods, achieving an $R^2 = 0.78$ and providing a more detailed representation of regional economic disparities.

For Vietnam, several studies have also used NTL data to analyze and predict economic indicators. Since official GDP data for Vietnam have only been available from 2010 onward, Suleiman et al. (2025) estimated Vietnam's subnational GDP from 1992 to 2010 using NTL data, demonstrating that remote sensing data can serve as a reliable predictor of Vietnam subnational GDP. In addition to that, Zhang (2025) demonstrated that NTL data can effectively capture spatial economic disparities patterns in Vietnam, particularly by identifying clusters of economically developed and underdeveloped regions. These findings suggest that the relationship between NTL data and economic activity observed in other regions also holds in Vietnam, providing a strong basis for applying NTL methods for GDP spatialization. However, to the best of my knowledge, no existing study has applied NTL data to spatialize Vietnam's GDP at the regional level, although similar applications have been conducted at the global level (Wang et al., 2019).

Therefore, this study aims to address this gap by adapting and extending the methods used in previous research to spatialize regional GDP in Vietnam using NPP-VIIRS NTL data. In addition, this research will develop an automated and reproducible analytical workflow that takes NPP-VIIRS imagery and regional GDP data as inputs and produces spatial GDP estimates.

This paper is organized as follows. Section 2 describes the data sources and general methodology used across all implemented models, including the data preprocessing procedures and the overall structure of all models. The results section provides a more detailed examination of the implementation specifications and findings for each model, along with the corresponding statistical analyses. The discussion section evaluates the strengths and limitations of each method and explains the observed outcomes. Finally, the conclusion summarizes the main findings of the study.

2 Materials and Methods

2.1 Study Area

In this study, I analyze the Gross Regional Domestic Product (GRDP) of provinces and centrally governed cities in Vietnam, which span latitudes from 8°10′N to 23°24′N and longitudes from 102°09′E to 109°30′E. Vietnam has a total land area of approximately 331,212 km^2, of which mountains, plateaus, and hills account for approximately three-quarters. According to the National Statistics Office of Vietnam (NSO), Vietnam's GDP reached 12,847.6 billion VND (approximately USD 514 billion) in 2025, reflecting a growth rate of 5.67%[1] during the 2020–2025 period. Economic development across Vietnam's regions is highly uneven and the country's rapid economic growth has created significant regional disparities. Zhang (2025) documented substantial spatial heterogeneity in Vietnam's economic development: major urban centers such as Hanoi, Ho Chi Minh City, and Hai Phong experienced rapid growth, whereas regions such as the Northwest and the Mekong Delta developed at a comparatively slower pace.

An important consideration for this study is the recent change in Vietnam's provincial level administrative boundaries. On June 12, 2025, the Vietnamese National Assembly passed Resolution No. 202/2025/QH15, reducing the number of provincial level administrative units from 63 to 34, consisting of 28 provinces and 6 centrally governed municipalities. Therefore, this analysis will be adjusted according to the new boundaries.

2.2 Data

This study utilizes data from three sources. First, the primary indicator, *Gross Regional Domestic Product* (GRDP) per capita, is obtained from the Statistical Yearbook of Viet Nam 2025, covering the period from 2020 to 2025. The NSO also provides data on national GDP, GRDP by capita, population, and population density across regions of Vietnam. Second, NTL data are obtained from the *Earth Observation Group* (EOG) and are derived from VIIRS observations collected aboard the Suomi NPP satellite, part of the *Joint Polar Satellite System* (JPSS), covering the period from 2012 to 2025. Third, administrative boundary shapefiles are obtained through the Humanitarian Data Exchange portal, provided by the *United Nations Office for the Coordination of Humanitarian Affairs* (OCHA). According to the boundary dataset documentation, the boundary alignments incorporate the latest updates from official Vietnamese government decrees. For the commune-level boundary shapefile, I collect from gis.vn, a Vietnamese geospatial platform and community tool.

2.2.1 National Statistics Office of Vietnam

GRDP per capita data for the 34 provinces and centrally governed municipalities in Vietnam covering the period from 2020 to 2025 were obtained from the 2025 Statistical Yearbook. The yearbook is an annual publication issued by the National Statistics Office of Vietnam (NSO) and is available through the official NSO website. The GRDP figures reported in the 2025 edition are the first to be calculated according to Vietnam's new provincial administrative classification. These figures are reported at constant prices, with 2010 as the base year, and therefore adjusted for

[1]This number is calculated by using Vietnam's GDP from 2025 and 2017, where Vietnam's GDP is according to NSO.

inflation. In previous editions, GDP was reported according to the former provincial administrative classification. The GDP figures are expressed in constant local currency values.

For population data, the annual national statistical yearbooks provide population and population density estimates at the provincial level. In addition, the statistical yearbooks published by individual provinces provide population data at a finer geographic level, including the commune level. However, GRDP data are not available at the provincial and commune level. Therefore, population data can be incorporated into the analysis at both the provincial and commune levels.

2.2.2 Nighttime Light data preprocessing

The data is obtained in GeoTIFF format. Here, I will use the Annual VNL V2 dataset for 2020–2025 because it has already undergone processing to reduce the effects of outliers and temporary light sources (Elvidge et al., 2021). The NTL data is clipped to Vietnam's administrative boundary using the administrative boundary data. Then, I extracted the radiance value of each pixel. The radiance value, measured in $nW/cm^2/sr$, represents the intensity of nighttime illumination at each location and will serve as the primary predictor in the models. The pixel level NTL data will then be spatially aggregated to the corresponding administrative units at both the provincial and commune levels. These measures will then be joined with the corresponding province GDP for model training and evaluation. The pixel level NTL will also be used for spatialization of the GDP throughout Vietnam.

3 Methods

The procedure for spatializing GDP consists of two steps. First, I will develop and evaluate predictive models at the provincial level using NTL and population data. The model predictions will then be validated against observed provincial GRDP figures to compare alternative modeling approaches and identify the model that provides the most accurate estimates of provincial GDP. Second, after selecting the optimal model based on its provincial level predictive performance, I will apply the same model specification to NTL and population data at the commune level within each province. This allows the estimated provincial GDP to be further disaggregated among individual communes based on their observed NTL and population. In this way, the provincial level model serves as the basis for generating commune level GRDP estimates, providing a much finer spatial representation of economic activity than is available in standard statistical reports.

In order to evaluate the predictive power of these models, I will use the temporal holdout approach. Data from 2020 to 2024 will be used as the training set, while data from 2025 will be reserved as the testing set. The model will be trained using the training dataset, and then be applied to the previously unseen 2025 data. Its predictive performance will be evaluated using R-squared and MSE by comparing the predicted provincial GDP values with the reported 2025 GRDP values. Because the 2025 data are not used during model training or hyperparameter selection, this temporal holdout approach provides an independent assessment of how well the models generalize to a new year and helps reduce the risk of overfitting.

The predictive models use three explanatory variables: the mean nighttime light radiance within each region, the population of the corresponding region, and the number of pixels representing its geographic area. The number of pixels is used as a proxy for regional area rather than relying on officially reported area figures, which may contain measurement inaccuracies or inconsistencies.

Although pixel counts may also introduce some degree of measurement error, they provide a more consistent and spatially representative approximation of regional area in the context of this study.

3.1 Predictive models

For convenience, we refer to the national, provincial, and commune levels as Level 0, Level 1, and Level 2, respectively. For each model, define the following notation:

- $y \in \mathbb{R}^n$: the vector of observed outcomes, representing GRDP per capita.
- $x_i \in \mathbb{R}^3$: the vector of standardized explanatory variables for observation i, comprising `mean_radiance`, `population`, and `pixel_count`.
- P_i: the population associated with observation i at the corresponding administrative level.
- $a \in \mathbb{R}$: the specified aggregate GDP or GRDP at the corresponding administrative level.

3.1.1 Ordinary Least Squares

The linear model is formulated as the following constrained optimization problem:

$$\begin{aligned} \min_{\alpha,\beta} \quad & \|y - \hat{y}\|_2^2 \\ \text{subject to} \quad & \sum_{i:\,\text{year}_i = t} P_i \hat{y}_i = a_t, \quad t \in \{2020, \ldots, 2025\}, \\ & \hat{y}_i = \alpha_{\text{year}_i} + x_i^\top \beta. \end{aligned}$$

The objective function is identical to that of ordinary least squares (OLS), minimizing the sum of squared differences between observed and predicted GRDP per capita. However, I introduce additional equality constraints to ensure that the predictions are consistent with the reported aggregate economic output at the national level for each year. That is, the sum of products of population and predicted GRDP per capita must equal the corresponding reported aggregate GDP, as enforced by the first constraint.

For the level 2 predictions, I apply the following adjustment to the model predictions to ensure consistency with the reported provincial GRDP per capita:

$$\tilde{y}_{ij} = \hat{y}_{ij} + \delta_j,$$

where

$$\delta_j = \frac{g_j N_j - \sum_{i \in j} P_{ij} \hat{y}_{ij}}{\sum_{i \in j} P_{ij}}.$$

Here, δ_j is the common adjustment term for every level 2 prediction within province j, g_j denotes the reported GRDP per capita of province j in 2025, N_j is the province's reported total population, and P_{ij} is the population of commune i within province j. This adjustment distributes the difference between the reported provincial GRDP and the aggregate implied by the original commune-level predictions proportionally to population, while preserving the relative differences among the original predictions.

Consequently, the adjusted predictions satisfy the following aggregation constraint for each province j:

$$\sum_{i \in j} P_{ij} \tilde{y}_{ij} = g_j N_j.$$

3.1.2 Ridge Regression

The Ridge Regression model is formulated as the following constrained optimization problem:

$$\begin{aligned} \min_{\alpha,\beta} \quad & \|y - \hat{y}\|_2^2 + \lambda \|\beta\|_2^2 \\ \text{s.t.} \quad & \sum_{i:\, \text{year}_i = t} P_i \hat{y}_i = a_t, \quad t \in \{2020, \ldots, 2024\}, \\ & \hat{y}_{it} = \alpha_t + x_i^\top \beta, \quad t \in \{2020, \ldots, 2024\}, \\ & \lambda \geq 0. \end{aligned}$$

The objective function follows the standard Ridge Regression formulation, combining the ordinary least squares loss with an L_2 regularization penalty. The tuning parameter λ controls the strength of this penalty, helping reduce the impact of multicollinearity among the explanatory variables, particularly the correlation between population and mean nighttime light radiance. The optimal λ is selected through a grid search over candidate values, with the value yielding the lowest RMSE on the 2025 holdout dataset chosen for the final model.

Since the 2025 year-specific intercept cannot be estimated directly from the training observations, it is extrapolated from the estimated 2020-2024 local intercepts using linear temporal extrapolation weights:

$$v_t = \frac{1}{T} + \frac{(2025 - \bar{t})(t - \bar{t})}{\sum_{j=1}^{T} (t_j - \bar{t})^2}.$$

where T is the number of training years, t represents an individual training year, and $\bar{t}$ is the mean training year. The extrapolated intercept is then:

For Level 2 predictions, the model applies the same adjustment term, δ_j, as in the Ordinary Least Squares approach. This adjustment serves the same purpose of ensuring that the population-weighted aggregate of the predicted GRDP per capita values is consistent with the corresponding province-level GRDP.

3.1.3 Geographically Weighted Regression. (GWR)

The GWR model assumes that observations closer to each other geographically tend to have more similar relationships between the explanatory variables and the dependent variable. Therefore, unlike previous models, GWR allows regression coefficients to vary across geographic locations to account for spatial heterogeneity.

The model is expressed as:

$$y_i = \alpha_{t_i}(u_i, v_i) + \sum_{k=1}^{p} \beta_k(u_i, v_i) x_{i,k} + \varepsilon_i,$$

where (u_i, v_i) represents the geographic coordinates of observation i, and the intercept and regression coefficients vary across locations. In this study, the model includes three explanatory variables: mean nighttime light radiance, population, and pixel count, along with year-specific intercepts. The geographic coordinates used in this implementation correspond to the centroids of the respective regions, which serve as the locations at which the spatially varying coefficients are estimated.

The models are implemented using the `GWmodel` package in R (Gollini et al., 2015). Additionally, I also implement the CVXR-based local regression model is implemented with an L_2 regularization penalty on the explanatory-variable coefficients:

$$\min_{\boldsymbol{\alpha}(s),\boldsymbol{\beta}(s)} \sum_{i=1}^{n} w_i(s) \left[y_i - \alpha_{t_i}(s) - \mathbf{x}_i^\top \boldsymbol{\beta}(s) \right]^2 + \lambda \|\boldsymbol{\beta}(s)\|_2^2.$$

The parameter λ controls the strength of regularization, while the year-specific intercepts remain unpenalized. Since the 2025 year effect cannot be estimated directly from the training data, it is extrapolated using the same approach as in the Ridge Regression model. The adaptive bandwidth and, for the CVXR model, the regularization parameter are selected based on the RMSE on the 2025 holdout dataset. The selected models are then refitted using all available years to generate Level 2 predictions.

3.1.4 Neural Network

3.2 GDP spatialization

After selecting the optimal models and their corresponding parameters, the final models will be retrained using the complete available training dataset. These models will then be used to estimate GDP at the commune level. This disaggregation treats each province as the higher-level geographic unit and its constituent communes as the lower-level units. Commune-level NTL, population, and other model inputs will be used to estimate the relative economic activity of each commune within its province.

An important assumption is that the disaggregation from national level down to provincial level is similar to the disaggregation from provincial level to commune level. To ensure consistency between geographic levels, the commune-level estimates will be subject to an aggregation constraint such that the sum of the estimated GDP across all communes within a province equals the corresponding provincial GDP. This constraint ensures that the resulting GDP estimates are spatially disaggregated estimates of provincial GDP rather than unconstrained independent predictions, while maintaining consistency with the higher-level provincial estimates.

4 Results

4.1 Statistical analysis

During the training processes, we use the 2025 holdout to compute the R^2 and the $RMSE$. Table 1 shows the R^2 and the $RMSE$ of each of the implemented model on the 2025

We can see that imposing the aggregation constraints has a negligible impact on predictive performance. Compared with OLS, the constrained regression increases the RMSE by only 0.17%, while R^2 decreases slightly from 0.4730 to 0.4712. Thus, the different between the constrained and the unconstrained model are minimal

Table 1: Model Performance Comparison

Model	R^2	RMSE
Ordinary Least Squares	0.37315	35.95181
Constrained Linear Regression	0.38352	35.65298
Constrained Ridge Regression	0.38499	35.61068
CVXR local regression	0.85426	17.33498
GWR	0.43612	34.09816
Neural Network	0.83270	18.57318

4.2 Predictions at a finer spatial scale

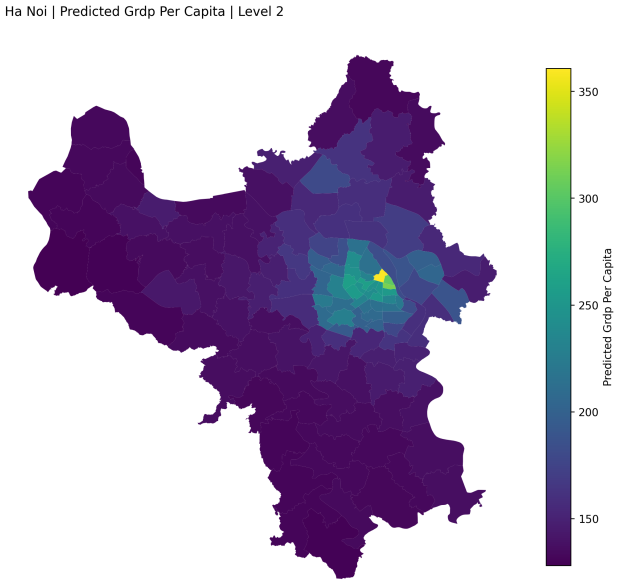


Figure 1: Predicted GRDP per capita at the commune level in Ha Noi

Figure 1 shows the estimated GRDP per capita at level 2 in Ha Noi. The spatialization results reveal a plausible geographic pattern, with higher predicted GRDP per capita concentrated in the city center. This pattern is consistent with the concentration of economic activities in urban areas.

However, this is the only good predictions. When looking at the general

4.3 Ridge Regression Model

To select the regularization parameter λ, I evaluate model performance across a grid of values, using the temporal holdout approach. As shown in Figure 2, the best predictive performance occurs when $\lambda = 15.85$. This suggests that penalizing the regression coefficients does improve predictive accuracy in this application, but minimally.

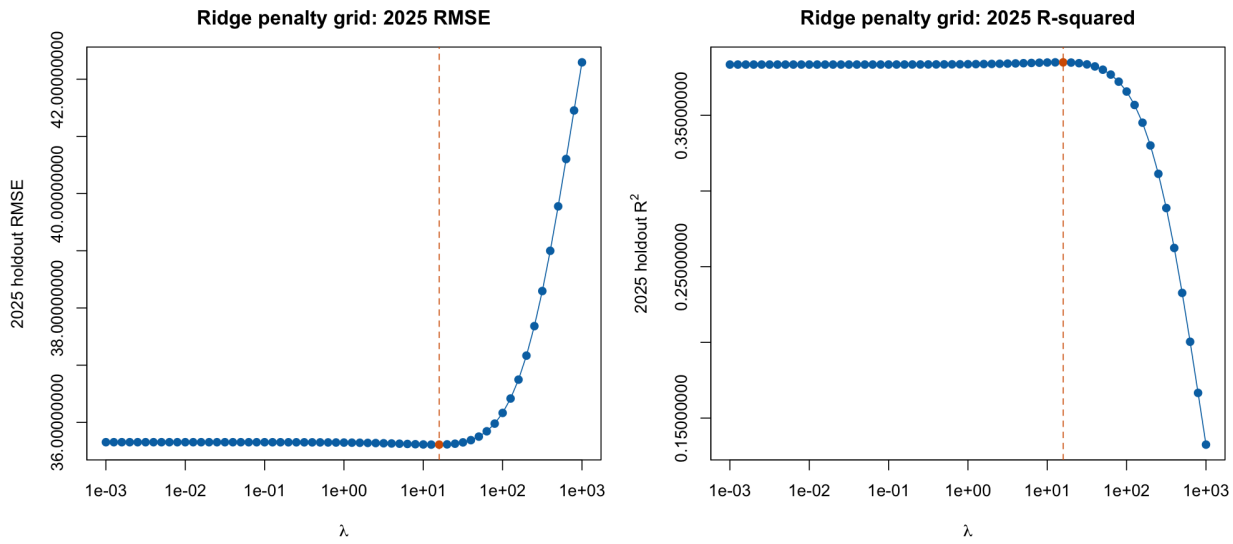


Figure 2: RMSE and R^2 across candidate values of the ridge regularization parameter λ.

5 Discussion

A general observation is that more complex models tend to overfit the Level 1 data and may not generalize well when applied to Level 2 predictions. Therefore, even when these models achieve relatively high R^2 and low RMSE on Level 1, their performance does not necessarily translate into reliable Level 2 estimates. This suggests that evaluating models solely based on their Level 1 predictive performance is insufficient for determining their suitability for spatial disaggregation.

Another important consideration is the difference in the number and characteristics of observations across geographic levels. The models are trained and evaluated using provincial observations, but their final application is at the commune level. Relationships learned from provincial data may not remain consistent at a finer geographic scale. This limitation is particularly important for more complex models, which may capture patterns specific to the provincial training data rather than relationships that generalize across smaller regions.

6 Conclusion

This study compared several predictive approaches, including Ordinary Least Squares, Ridge Regression, Geographically Weighted Regression, and Neural Networks, to investigate the relationship between nighttime light radiance and GRDP per capita across provinces in Vietnam. Using mean nighttime light radiance, population, and pixel count as explanatory variables, the study examined the ability of these models to estimate regional GRDP per capita and extend the predictions to the commune level.

The findings is that strong predictive performance at the provincial level does not necessarily imply reliable commune-level estimates. More complex models may capture patterns in the provincial data that do not generalize to smaller geographic units, even when aggregate constraints are imposed. Therefore, the suitability of a model for spatial GRDP estimation should be assessed not

only through conventional statistical metrics but also through the plausibility and consistency of its predictions across geographic scales.

Despite these limitations, this study provides a framework for using nighttime light imagery and statistical learning methods to investigate the spatial distribution of economic activity in Vietnam. Future work could improve the reliability of commune-level estimates by incorporating additional regional information, exploring methods that explicitly account for differences across geographic scales, and evaluating the resulting estimates against more detailed local economic data.

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